7 Fig. 7.1 shows a barrier at the entrance to a car park. The wooden barrier arm has a weight of 60 N which acts through the centre of gravity at the position shown on Fig. 7.1.

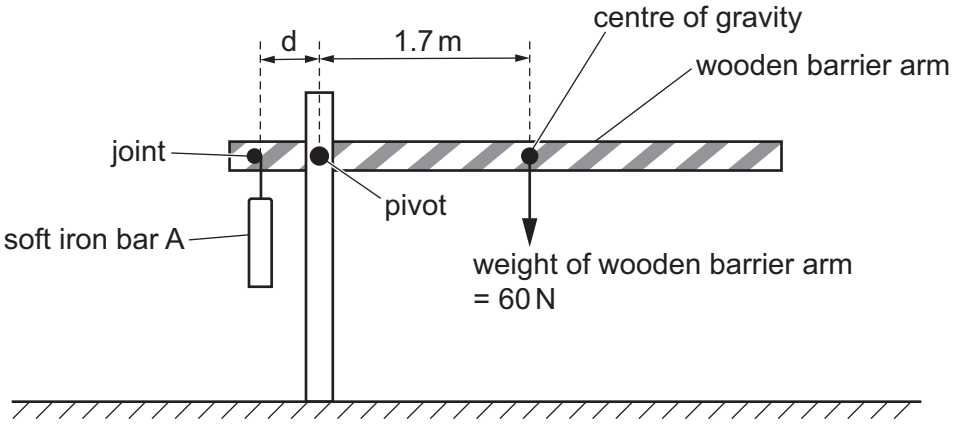

Fig. 7.1

(a) Initially the wooden barrier arm is horizontal.

(i) Using Fig. 7.1, calculate the clockwise moment of the weight of the wooden arm about the pivot.

clockwise moment = ... Nm [1]

(ii) The wooden barrier arm is in equilibrium. The mass of the soft iron bar A is 23 kg.

Calculate the distance d between the pivot and the joint holding the soft iron bar A.

distance d = ... [3]

DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN
DO NOT WRITE IN THIS MARGIN

(b) Fig. 7.2 shows a coil attached to a power supply placed below the soft iron bar A.

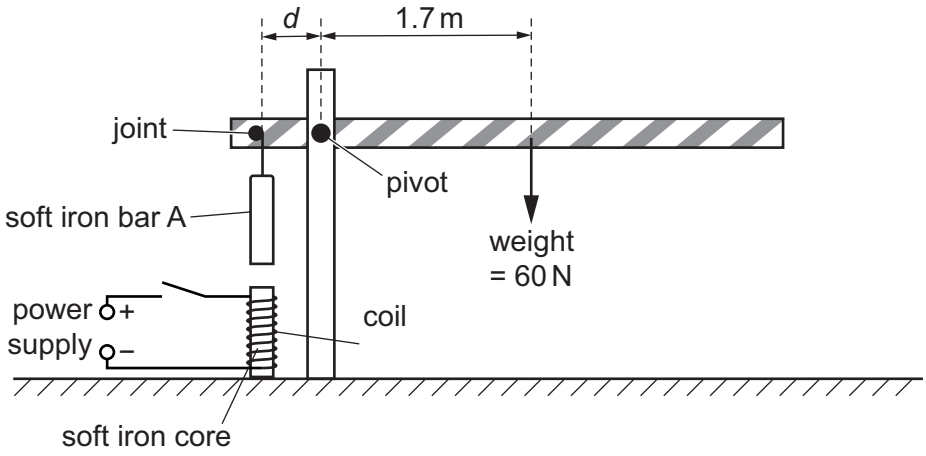

Fig. 7.2

(i) State and explain what happens to the wooden barrier arm when the switch in the coil circuit is closed.

statement .....

explanation .....

.....

.....

.....

[3]

(ii) The switch is opened. An operator decreases the potential difference across the coil and the switch is closed.

State and explain how the effect on the wooden barrier arm compares with the effect in **(b)(i)**.

statement .....

explanation .....

.....

[2]

(iii) A student suggests that the soft iron bar A is replaced by a steel bar. Explain why a steel bar is less effective than a soft iron bar in the barrier.

.....

.....

..... [2]

[Total: 11]

DO NOT WRITE IN THIS MARGIN